

# Akshaya Thenkarai Lakshminarasimhan

San Diego, CA | athenkarailakshminar@ucsd.edu | +1 (858)518-4008 | portfolio-akshaya.netlify.app/  
linkedin.com/in/akshaya-tl/ | github.com/Akshaya-TL | kaggle.com/phoenix2001  
https://scholar.google.com/citations?user=9WqDftEAAAAJ&hl=en

## Education

**University of California, San Diego** Master of Science in Computer Science Sept 2024 – Mar 2026

- **Coursework:** CSE 291 H00: Advanced Data Driven Text Mining, CSE 258: Recommender Systems and Web Mining, CSE 256: Statistical Natural Language Processing, CSE 253: Data Systems for ML, CSE 291 J: Fairness, Bias and Transparency in ML, CSE 202 Algorithm Design and Analysis
- **Research Assistant** in **Lab 1:** McAuley Lab (Recommendation System Lab) **Lab 2:** STABLE Labs (Jishen Zhao, ML Systems Lab)
- **Teaching Assistant** CSE 255: Big Data Analytics using Spark - Prof Yoav Freund
- **Part-Time Web Developer** Part-time Web Developer ITA Conference website for Prof. Alon Orlitsky

## Publication

Abeer Alessa, Akshaya Lakshminarasimhan, Param Somane, Julian Skirzynski, Julian McAuley, Jessica Echterhoff. **How Much Content Do LLMs Generate That Induces Cognitive Bias in Users? (IJCNLP AACL 2025 (Main Conference) accepted)**

TL. Akshaya, K. Swetha, S. Pravalika, U. Chandrasekhar. **"Segmenting MRI Images Using Federated Learning For Brain Tumor Detection"** in International Conference for Machine Learning and Big Data Analytics - **ICMLBDA Conference 2023, Best Paper Award in Technical Session for Data Science, Big Data, and Data Mining for Machine Learning**, Scopus Indexed Cham: Springer

## Work Experience

**Amazon Web Services** Jun 2025 – Sept 2025  
Software Development Engineer Intern

- Part of the **Amazon Web Services EC2 AMI Platform team**, contributing to the development and enhancement of AWS AMI.
- Developed an application **automating Legacy BIOS AMI to UEFI conversion**, reducing customer effort from several hours of manual work to **52 minutes** for a **30 GB OS**, enabling **100% automation** and eliminating tedious manual partitioning and reconfiguration steps.
- **Solved critical challenges** by enabling **Extensible Primary Partition**, **careful transformation of Root Volume** and **retention of Billing Code** in the resultant UEFI AMI.
- Added **feature assortment** support to include **UEFI-only mode**, **TPM integration**, and **Secure Boot enablement** to **enhance EC2 instance security and malware protection** with **zero post-conversion configuration** required.
- Achieved **100% data integrity** in customer AMI transformation, **preserving billing codes** and **metadata**
- Authored the system design document, proposed architectural solutions, and implemented the application with **high-quality, modular code**, supported by comprehensive unit and integration testing.
- **Tech Stack:** AWS EC2, Lambda, Step Functions, EBS, SSM, IAM, CDK, Python, TypeScript, PowerShell, Git.

**Drona Pay** Mar 2024 – Aug 2024  
System Analyst

- **Spearheaded the development of Behavioral Risk Analysis Scorer**, which analyzes diverse data points performing real-time fraud prevention, achieving detection and prevention **under 30 milliseconds**, contributing towards building a secure, next-gen payment and lending platform.
- **Streamlined Spring Boot based backend-service** by modularizing the core business logic, integrating JsonLogic Library features, enhancing the error handling mechanisms and ability to turn on/off the Google Protocol Buffer features to stream the data into Apache Kafka. This reduced boilerplate code and enabled dynamic transformations, enhancing the risk scoring process.
- **Enhanced orchestration pipelines** developed in Apache Airflow, improving system efficiency and reliability.
- **Tech stack:** Java, Python, Spring Boot, Maven, PostgreSQL, Apache Kafka, JsonLogic, Google Protocol Buffers, Airflow, Redis, Docker, Postman, ArgoCD, Version Control Git, Agile Methodologies, JIRA

**Experian** Aug 2022 – Oct 2023  
Software Engineer Trainee

- **Contributed towards a Big Data project:** CensusHub whose purpose is to ingest millions of records of data from various data providers and enable quick turn-around-time for generating Experian Verify Reports.
- **Led migration of CensusHub Big Data Project** to a high-availability, fault-tolerant system using Apache Airflow and AWS services, improving system robustness and performance.
- **Optimized file processing time by 98%** (reducing from **16 hours to 15 minutes**) for large datasets of nearly **4GB** of data (**nearly 2 million records** of data). Additionally, troubleshooted and resolved issues causing inconsistent processing times in AWS EMR clusters handling Spark jobs.
- **Detected data inconsistencies and developed a solution to preserve data integrity** in the data warehouse, ensuring accurate storage of information from diverse international data providers.
- **Contributed to the micro-service up-gradation** in EVAPI: core system for the generating income and employment reports for our

- customers. Upgraded the services' Java (8 to 17) and Spring Boot versions (2.x to 3.0), achieving **100%** Veracode compliance, fixing Sonar issues, reducing cognitive complexity, performing code refactoring and improving code security.
- **Contributed to full-stack application** using Angular and NodeJS which involved vulnerability issue fixes and enhancements.
- **Participated in Internal Hackathon** building an **NLP-Based Bot**: Experian CIRA for customer incident handling
- Developed an Automated testing with Robot Framework which helped in quick development and deployment process.
- **Tech Stack**: Java, Python, Spring Boot, Gradle, PostgreSQL, Apache Airflow, AWS - S3, EC2, Lambda, EMR, SecretsManager, Angular, NodeJS, Postman, Jenkins, Veracode, SonarQube, Version Control Git, JIRA, Agile Methodologies

Target

Feb 2022 – Oct 2022

Data Science Trainee

- Selected for the Target UpCurve Data Science Cohort, **among top 92 women candidates** across India.
- **Worked on Planogram Optimization (PLUTO) project**, analyzing product data to optimize product placement for maximum sales.
- **Developed machine learning and deep learning models** to predict sales in dollars/units, that identify the most optimised product placements, achieving a **highest accuracy of 95.4%**.

Achievements

- 🏆 **Best Research Paper Award in ICMLBDA Conference** conducted by NIT Arunachal Pradesh in July, 2023 for the project titled Segmenting MRI Images Using Federated Learning For Brain Tumor Detection
- 🏆 **TCS Codevita Season 10 - global rank 626 in Semi-Finale** round participated by 10 Lakh+ coders across the globe.
- 🏆 **Certificate of Distinction** for completing the course "Artificial Intelligence for All" taught by Dr. Rao Vemuri, Dean Emeritus University of California Davis, as part of Indo Universal Collaboration for Engineering Education (IUCEE)
- 🏆 Among the **Top 92** candidates selected for **Target UpCurve Data Science Cohort, 2022** out of 1000+ applicants all over India.
- 🏆 Secured a rank of **324 among 2804** participants in **Get A Room: ML Hackathon** achieving an accuracy of **70.013%**

Technical Skills

- Programming Languages**: Python, C++, Java, C, Haskell, Typescript, Go
- Big Data & Data Processing**: Apache Airflow, Apache Kafka, JsonLogic Library, Google Protocol Buffers
- Web & Application**: Spring Boot, Angular (MEAN stack), Flask, Maven, Gradle, HTML, JavaScript, BabylonJS
- Cloud Services**: Amazon Web Services (AWS): Step Functions, EBS, S3, EC2, EMR, Lambda, SecretsManager
- AI/ML**: NumPy, Pandas, ScikitLearn, Matplotlib, Keras, Tensorflow, PyTorch, OpenCV, Federated Learning, HuggingFace, LLM, Deep Learning, Computer Vision, Jupyter, Semantic segmentation
- Databases & Version Control**: MySQL, PostgreSQL, git, agile methodologies, jira
- Developer Tools**: Postman, Docker, SonarQube, Veracode, Google Collab, Jupyter Notebook, VS Code, PyCharm, IntelliJ,

Projects - Software Development & AI/ML

Captions to Cart: A cross domain recommendation system | AI/ML

Link

- Implemented a cross-domain recommendation system that leverages Instagram post content to suggest relevant Amazon products by utilizing TF-IDF and cosine similarity

Self-Improving Zero-Shot BioNER | AI/ML

Link

- We present a self-improving zero-shot approach for biomedical NER that combines Dictionary-Infused RAG, adaptive consistency, and self-evaluation guided beam search.

Social Sense: Decoding Digital Emotions Across Meta, Instagram, and Twitter | AI/ML

Link

- Aims to decode and classify emotions expressed in social media posts across Facebook, Instagram, and Twitter by leveraging various machine learning techniques, from traditional approaches to state-of-the-art deep learning models.

Planogram Optimisation (PLUTO) | AI/ML

Link

- Developed Machine Learning and Deep Learning models for optimizing the placements and numbers of facings of each item on a planogram with highest accuracy of **95.4%**.

2D Shape Extrusion & Manipulation | Software Development

Link

- Created a Babylon.js application that allows the user to draw arbitrary 2D shapes on the ground plane, extrude them into 3D objects, and then manipulate those objects by moving the object and editing their vertices using buttons for mode selection & visual cues.

Fresh From Farm Website | Software Development

Link

- A MEAN-stack-based E-commerce web application that helps connect customers and farmers directly.

Langton's Ant - A Universal Turing Machine Application | Software Development

Link

- A 2-dimensional UTM application that involves concepts of Formal Language and Automata theory

Semantic Segmentation for Autonomous Driving | AI/ML

Link

- Built a Deep learning system that performs pixel-by-pixel image segmentation from the camera feed to enable Autonomous driving.